Inventory Objective Certificates and Physical Fleet Realization

A research section accompanying the stable GI-TBC method

1 What a tight inventory relaxation establishes

Let D be a qualified integer-inventory relaxation of the complete improving physical domain. With the original coefficients and normalization, write

$$r_i = c_i Y_i, \quad S = \sum_i r_i, \quad A = \sum_{i<j} |r_i - r_j|, \quad H = \sum_i \omega_i |r_i - 1|, \quad J = G + H,$$

where $G = A/(nS)$ for $S > 0$, with the original zero convention otherwise. The coefficient ω_i already includes the objective weight; it is not multiplied by that weight again. For the root scopes considered here, the pricing epigraph has $\gamma_L = 0$, so its original inventory objective is J. Denote physical realizability with the original time limit and fleet by $R(Y)$, and define

$$F_I = \min_{Y \in D} J(Y), \qquad F_P = \min_{Y \in D : R(Y)} J(Y).$$

Thus $F_I \le F_P$. The integer-inventory pricing and proof-consumption studies refine F_I and its scope-valid contribution to the complete GI-TBC ledger. Their inventory upper bounds do not supply vehicle routes.

This separates the roles of structural relaxation and physical search. Integer inventory states, normalized ratio products and qualified pricing can strengthen the original objective relaxation while leaving routing unresolved. A root inventory lower that covers the entire improving domain can supplement the complete search's physical lower in the proof ledger. The search may continue to use its own bounds for adaptive-mass decisions: a stronger proof floor need not change interval branching, native rows or incumbent production. Conditional child floors instead require complete child coverage and their corresponding local scopes. These proof-placement choices are separate from whether a good inventory can be turned into a route.

Proposition 1. *Suppose $L_I \le F_I \le J(\widehat{Y}) = U_I$. If an inventory with objective U_I has a verified complete physical fleet, then $L_I \le F_P \le U_I$ and $0 \le F_P - F_I \le U_I - L_I$.*

Proof. Restricting the inventory domain to physically realizable vectors cannot lower its minimum. The checked fleet supplies a member of that restricted domain with value U_I. Combining the two inequalities gives the result. □

The conclusion is about the recorded domain. If D is an improving domain, its incumbent-cutoff complement must also be covered before a global certificate is asserted. A pricing tolerance of 10^{-4} is not substituted for the original complete-method certification tolerance. In particular, infeasibility of $\widehat{Y}$ alone does not imply $F_P > U_I$: other inventories can have the same or slightly greater price.

2 Objective invariance with spatial freedom

Partition stations by identical actual coefficients (c_i, ω_i). For a class C, let m_{Cv} be the multiplicity of inventory value v in $\widehat{Y}$. The class family is encoded by

$$\sum_v z_{iv} = 1, \qquad Y_i = \sum_v v z_{iv}, \qquad \sum_{i \in C} z_{iv} = m_{Cv}, \qquad z_{iv} \in \{0, 1\},$$

using only each original station's permitted support. The original hard inventory total and all physical conditions retain their meaning.

Proposition 2. *Every member of this family has the same S, A, G, H, J and unweighted inventory total as $\widehat{Y}$.*

Proof. Within a class, c_i is constant, so preserving inventory multiplicities preserves its ratio multiset. Taking the union across classes preserves the global ratio multiset, hence S, A, G. Constant ω_i also preserves the class sum of absolute deviations from one. Inventory multiplicities directly preserve the unweighted total. □

Initial inventories and geographical locations need not coincide within a class. Consequently, the required pickups, deliveries and routing difficulty can change. This is a practical use of objective invariance, not a new general symmetry or Benders theorem. Coefficients are compared by their exact binary representations, without epsilon grouping. The preselected best RX35 family has six legal members and total pickup 56–57; the best NB35 family is a singleton. This is a small finite opportunity, not an established large symmetry space.

3 A complete free-fleet physical query

The physical projection retains empty departures, at most one nonzero one-direction service per station, free vehicle attribution and route order, original heterogeneous capacities Q_k, and legal loaded returns. The depot's placeholder inventory supplies no bikes. Once the inventory family fixes J, the Gini and penalty objective formulation is unnecessary for the feasibility question.

For a selected state $v \neq b_i$, binary service assignments satisfy $\sum_k w_{kiv} = z_{iv}$. Only assignments with $|b_i - v| \leq Q_k$ are permitted. Their sums define each vehicle's station-service indicator and pickup/delivery quantities. Binary arc in/out degrees equal that indicator; depot departure equals return and is at most one. Standard ordering potentials eliminate disconnected station cycles. Instead of node-load big-M implications, arc loads satisfy

$$0 \leq f_{kij} \leq Q_k x_{kij}, \qquad f_{k0j} = 0, \qquad \sum_j f_{kij} - \sum_j f_{kji} = p_{ki} - d_{ki}.$$

At integer arcs, these equations give exactly the physical load sequence from empty departure. Return-arc load remains free within Q_k. Every real route lifts into these variables, and every integer solution decodes into a complete free fleet. This is the established inventory-time oracle projection with a standard arc-flow load formulation.

For each route, pickups P_k equal station deliveries D_k plus returned load R_k. The evaluator's handling time is therefore

$$t_p P_k + t_d(D_k + R_k) = (t_p + t_d)P_k.$$

The model charges this quantity plus travel against the original T. Total pickups can exceed Q_k when intermediate deliveries free capacity; only prefix loads are capacity-bounded. The time limit of an experiment is separate from the physical model's T.

Feasible status and an independently checked route are recorded separately. A truncated query without such a route is UNKNOWN. A qualified infeasibility result for the selector model covers exactly its finite family, not every inventory with that objective value.

4 Resource explanations and objective-level coverage

A necessary resource projection can preserve free station/vehicle attribution while replacing exact tours by lower bounds. Let $\ell(S)$ be a downward-rounded shortest depot tour through a small station subset S, computed from exact rational shortest paths of the actual travel coefficients. A route visiting S has travel at least $\ell(S)$, even if it also visits other stations. For vehicle service indicators s_{ki}, the following inequalities are necessary:

$$0 \le P_k - D_k \le Q_k, \qquad T^* \ge (t_p + t_d)P_k + \ell(S)\left(\sum_{i \in S} s_{ki} - |S| + 1\right).$$

Together with $T^* \ge (t_p + t_d)P_k$, a subset row is active only when all its stations are served by that vehicle. Singles, pairs and triples provide a finite, reproducible family of such conditions. A qualified resource lower greater than T can rule out its inventory condition; a feasible resource assignment is not a complete route.

To establish a physical objective gap, the exclusion must cover a whole level, for example all original physical solutions with $J(Y) \le \tau$. Infeasibility of a valid necessary relaxation of that entire level is sufficient; infeasibility of one inventory or one class family is not. A level-specific proof is consumed only in that level and its qualified coverage. If new exclusions delete an old inventory upper witness, it cannot remain an upper bound for a newly strengthened inventory problem. Logical no-goods and infeasibility learning retain their classical attribution [1, 2]; no replacement assignment master or unlimited learning loop is presumed here.

For a proposed complete enhancement, both the inventory and any physical completion must be generated and paid within that method's own run. An offline route establishes physical possibility, but its reuse would not measure an independently paid algorithm. At a common outer time origin, the absolute-gap difference decomposes as

$$(U_R - L_R) - (U_{E+} - L_{E+}) = (U_R - U_{E+}) - (L_R - L_{E+}).$$

Thus a reported improvement can be attributed to upper-bound production, lower-bound proof, or both without equating stronger inventory prices to faster routing. Unachieved certificate times remain undefined. Stable GI-TBC remains the reference while additional mechanisms are evaluated.

5 Evidence from near-optimal inventory states

The two prespecified roles use their original physical data: RX35 has three vehicles with capacities $(20, 25, 30)$ and $T = 6300$ seconds; NB35 has the same capacities and $T = 7200$ seconds. Their best saved inventory prices are approximately 0.23612171 and 0.39604655, respectively. The corresponding inherited physical uppers are 0.29966940 and 0.45029150. These inherited endpoints identify the unresolved physical opportunity; they are not fresh complete runs.

At a deterministic capacity of 4096 nodes, the exact fixed-inventory fleet models find no route and return UNKNOWN on both roles. Their measured outer costs are 1507.17 and 1285.90 seconds. In contrast, the necessary resource model proves a time lower of 6300.62843 seconds for the complete six-member RX35 family. After the inherited time-proof margin of 0.063 seconds, the lower remains

6300.56543, strictly greater than 6300. The original inventory and every permitted rearrangement are therefore impossible under the original time limit. This certificate costs 181.39 outer seconds. The difference in these costs compares two formulations; it does not isolate a speed benefit from class rearrangement.

For NB35, the resource call at 4096 nodes instead gives a guarded time lower of 6993.96882 seconds, below 7200. Its relaxed assignment upper of 7245.92316 seconds is not a fleet-time upper, because it has no complete tour. The singleton inventory remains UNKNOWN. Neither this unresolved query nor the RX35 family certificate changes a global original-objective bound by itself.

The resource obstruction has a concrete physical interpretation. Every member of the RX35 best family preserves total station inventory equal to the initial total. Since vehicles start empty, all return loads are zero and each vehicle's pickups equal its deliveries. The maximum individual station service is seven bikes, below every vehicle capacity, so the simple individual-assignment capacity filter is inactive. The exclusion already follows from per-vehicle balance, handling and the small-subset travel inequalities, before exact route order and prefix loads are imposed. Conversely, a separate metric tour lower combined with minimum aggregate handling gives 17793.65 seconds, below the fleet budget 18900. Aggregate handling and travel alone do not prove this family impossible.

To test propagation beyond a family, the same necessary conditions were appended to the complete original inventory domain, preserving the normalized objective formulation. An unfixed valid physical-route start and the qualified prior inventory lower were supplied. The revised diagnostic was stopped after 4099.31 outer seconds with 773 observed nodes: its reported global bound still remained at the prior floor and its node cost was rising. This was an early evidence-based reallocation before a proposed 1024-node checkpoint, not exhaustion of the 16384-node capacity. The earlier cold attempt cost 256.52 seconds and was stopped for the documented information-use revision. Both incomplete attempts are charged and preserved; neither yields a new objective certificate.

The preselected second RX35 family, with 20 legal members and original price approximately 0.23654185, is also excluded by the same necessary model. Its Native time lower 6300.45286 remains above 6300 after the 0.063 margin; the outer cost is 190.00 seconds. Both complete families, 26 vectors in total, can therefore be removed in a single finite counterfactual pricing batch. A nonexcluded saved stock of price 0.23685401 supplies a checked upper for that retained inventory domain, without asserting physical feasibility. This finite exclusion alone cannot raise the old physical proof lower by more than 0.00080191, about 1.26% of the inherited unresolved physical gap. This is a ceiling for that particular batch, not a limit on other resource conditions or route constructions.

Family	Members	Guarded time lower (s)	Original-T conclusion
RX35, best	6	6300.56543	Infeasible ($T = 6300$)
RX35, second	20	6300.38986	Infeasible ($T = 6300$)
NB35, best	1	6993.96882	Unknown ($T = 7200$)

Complete legal fleets constructed with time relaxed give genuine, albeit loose, upper bounds on the best-family minimum required time: 20647.93 seconds for RX35 and 24384.99 for NB35. The original physical reader independently validates their services and loads. These are time uppers, not original-T objective incumbents.

The single finite exclusion batch is repriced with the unchanged original RN objective at 16384 nodes. It costs 200.85 outer seconds and returns the same qualified lower 0.2360521029074557. The selected complete objective level 0.23648786033514127 remains unresolved. Thus the 26-vector impossibility proof has not established a strict gap between the global inventory and physical optima.

The two resource proofs plus repricing cost 572.25 seconds before qualification, without a new physical upper or global lower. Together with the whole-domain diagnostic and the small finite-batch opportunity ceiling, this provides no sufficiently promising uniform completion mechanism for a fresh complete algorithm. No complete enhancement is frozen and the conditional eight-arm panel is not run. Stable GI-TBC remains the default. The family obstruction is a mathematical result; more general low-price physical realization and objective-level exclusion remain open computational questions.

References

[1] J. N. Hooker and G. Ottosson. Logic-based Benders decomposition. *Mathematical Programming*, 96:33–60, 2003. doi:10.1007/s10107-003-0375-9.

[2] G. Codato and M. Fischetti. Combinatorial Benders' cuts for mixed-integer linear programming. *Operations Research*, 54(4):756–766, 2006. doi:10.1287/opre.1060.0286.